

# DiaTool-DPO

## Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

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ML Researchers & NLP Engineers | Tool-Augmented LLMs | RLHF Alignment

A novel DPO-based method that models TA-LLM interactions as an MDP with 5 dialogue states and 3 query types, automatically constructing preference pairs to improve slot-filling and tool call rejection without human labeling.

# SFT Alone Fails to Teach Slot-Filling and Tool Rejection

## The Core Problem

Tool-Augmented LLMs (TA-LLMs) must master three conversational skills:

- Ask follow-up questions for missing arguments
- Make accurate tool calls when arguments are complete
- Reject out-of-scope tool requests

SFT-only training with expert trajectories fails to teach these skills effectively, leading to hallucinated tool invocations and poor dialogue flow.

## Three Failure Modes of SFT-Only TA-LLMs

1

### Fails to Ask Follow-Up Questions

Model skips slot-filling QA and jumps to tool call with missing arguments, causing hallucinated or invalid invocations.

2

### Hallucinated Tool Invocations

Model makes tool calls without confirming required slots are complete, leading to downstream errors.

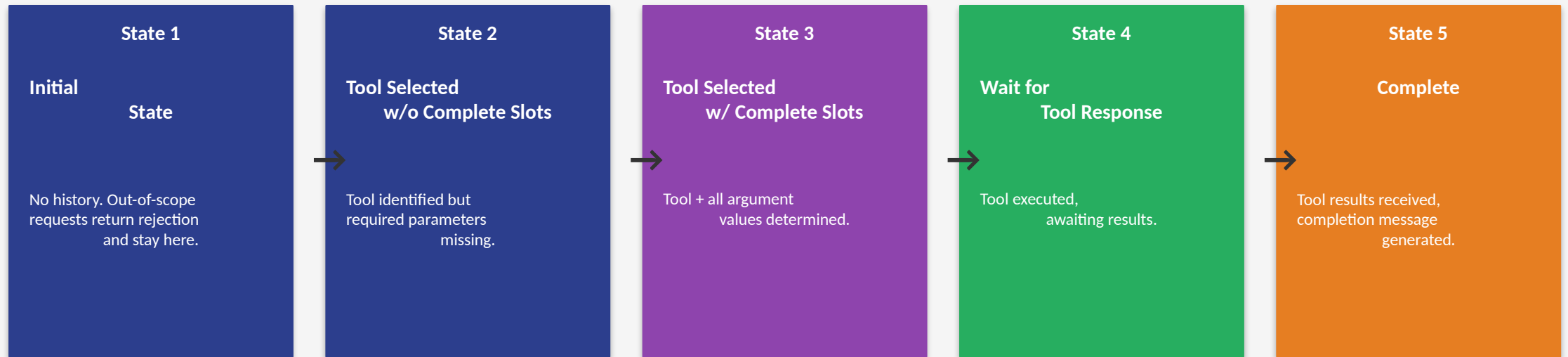
3

### Fails to Reject Inappropriate Tool Calls

Model attempts tool calls for unavailable functionality instead of gracefully rejecting the request.

→ DiaTool-DPO applies Direct Preference Optimization to automatically construct correct vs. incorrect dialogue trajectories, teaching slot-filling and tool rejection without human labels.

# Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



## Trajectory Paths by Query Type

|        |                                                                        |                                                                                       |
|--------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Type 1 | $1 \rightarrow 3 \rightarrow 4 \rightarrow 5$                          | Query contains all required arguments. No slot-filling needed. Direct tool selection. |
| Type 2 | $1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$ | Query missing arguments. Slot-filling QA loops N times until all fields gathered.     |
| Type 3 | $1 \rightarrow 1$                                                      | Out-of-scope request. Tool call must be rejected. Model stays in Initial state.       |

# Three Query Types Define Distinct State Trajectories

| Query Type | State Trajectory                                                       | Description                                                                     | Example                                                                                    |
|------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Type 1     | $1 \rightarrow 3 \rightarrow 4 \rightarrow 5$                          | All required arguments present in query. No slot-filling needed.                | "What is the weather in Seoul?" — Direct tool call; slots complete.                        |
| Type 2     | $1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$ | Missing arguments; N rounds of slot-filling QA until all fields gathered.       | "Book a flight to [unknown dest.]" — Model asks destination, then origin, then calls tool. |
| Type 3     | $1 \rightarrow 1$ (rejection)                                          | Requested functionality unavailable. Model must reject instead of calling tool. | "Call my dentist" — No dentist tool exists; model rejects gracefully.                      |

→ The 5-state MDP with 3 query types enables automatic construction of chosen/rejected trajectory pairs — no human labeling required.

# Automatic Paired Trajectory Construction Without Human Labels

## Pair Construction Strategy

### Chosen Trajectory

Correct dialogue flow:

- Ask all missing slots before calling tool
- Accept appropriate tool calls
- Reject out-of-scope requests

### Rejected Trajectory

Incorrect dialogue flow:

- Skip or hallucinate slots
- Call tool with missing arguments
- Accept inappropriate tool calls

## Dataset Composition (FunctionChat-Bench)

| Query | Chosen        | Rejected      | Lesson                      | Count | Diff. |
|-------|---------------|---------------|-----------------------------|-------|-------|
| T1    | 1→3→4→5       | 1→2→3→4→5     | Prevent redundant slot-fill | 2,089 | Easy  |
| T1    | 1→3→4→5       | 1→(2×N)→3→4→5 | Prevent redundant slot-fill | 562   | Hard  |
| T1    | 1→3→4→5       | 1→(2×M)→3→4→5 | Prevent redundant slot-fill | 2,530 | Hard  |
| T2    | 1→2→3→4→5     | 1→3→4→5       | Prevent slot hallucination  | 2,089 | Easy  |
| T2    | 1→(2×N)→3→4→5 | 1→3→4→5       | Prevent slot hallucination  | 562   | Hard  |
| T2    | 1→(2×N)→3→4→5 | 1→(2×M)→3→4→5 | Prevent slot hallucination  | 2,530 | Hard  |
| T3    | 1→1           | 1→3→4         | Tool call reject            | 567   | Hard  |
| T3    | 1→1           | 1→(2×N)→3→4   | Tool call reject            | 562   | Hard  |

Total: ~12,080 preference pairs | Mix of Easy & Hard trajectories | Language-agnostic (Korean + English)

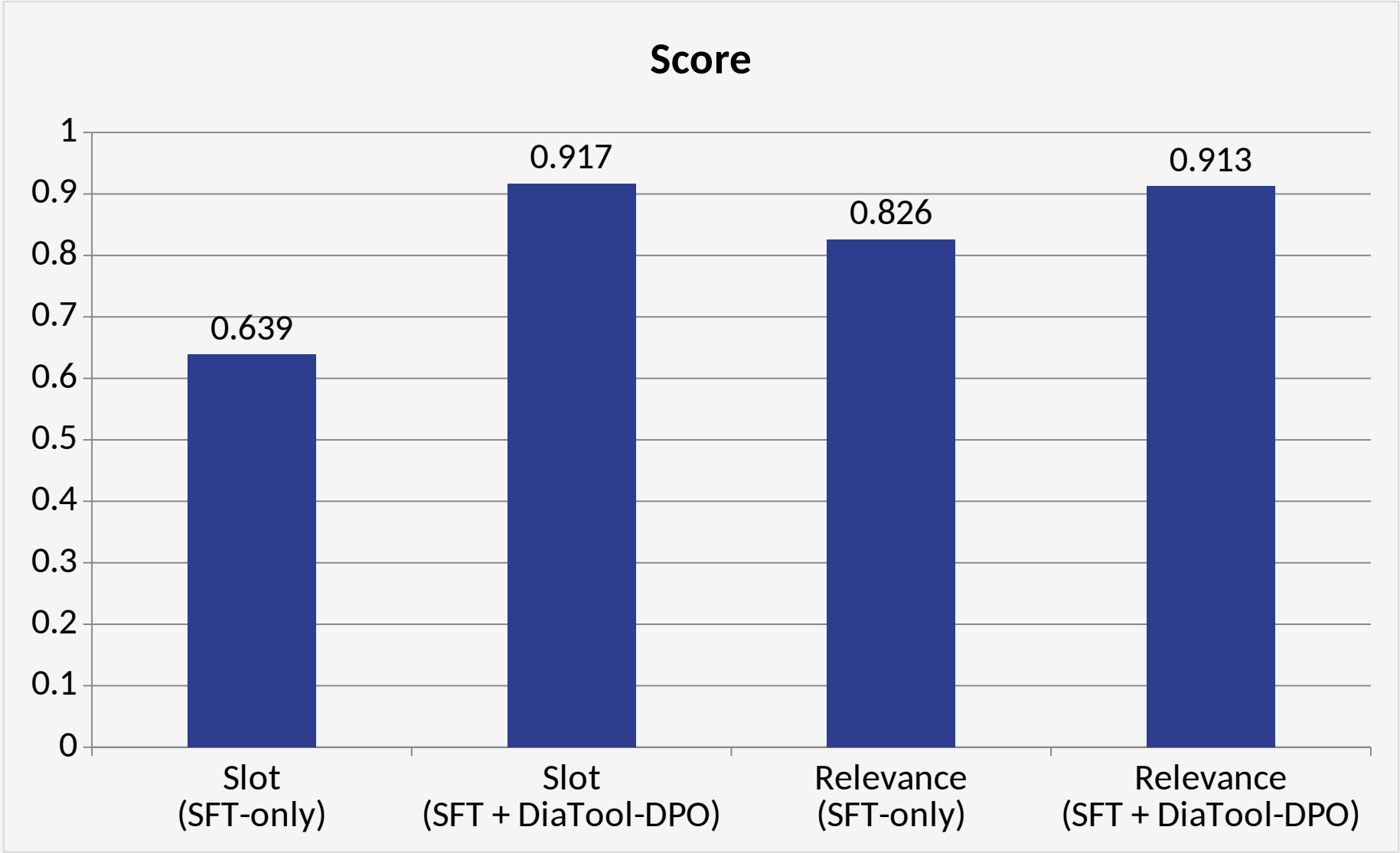
# SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

| Model      | Method            | Call  | Completion | Slot  | Relevance | Micro Avg | Macro Avg |
|------------|-------------------|-------|------------|-------|-----------|-----------|-----------|
| Prop.-8B   | DiaTool-DPO-only  | 0.314 | 0.700      | 0.833 | 0.609     | 0.575     | 0.614     |
| Prop.-8B   | SFT-only          | 0.900 | 0.916      | 0.694 | 0.913     | 0.870     | 0.856     |
| Prop.-8B   | SFT + DiaTool-DPO | 0.886 | 0.929      | 0.833 | 0.826     | 0.884     | 0.868     |
| Prop.-3.1B | DiaTool-DPO-only  | 0.357 | 0.551      | 0.528 | 0.391     | 0.455     | 0.457     |
| Prop.-3.1B | SFT-only          | 0.771 | 0.817      | 0.750 | 0.826     | 0.790     | 0.791     |
| Prop.-3.1B | SFT + DiaTool-DPO | 0.743 | 0.871      | 0.833 | 0.826     | 0.765     | 0.818     |
| LLaMA-3-8B | DiaTool-DPO-only  | 0.029 | 0.449      | 0.056 | 0.261     | 0.205     | 0.199     |
| LLaMA-3-8B | SFT-only          | 0.843 | 0.957      | 0.639 | 0.826     | 0.844     | 0.816     |
| LLaMA-3-8B | SFT + DiaTool-DPO | 0.857 | 0.929      | 0.917 | 0.913     | 0.905     | 0.904     |

## Key Observations

- SFT + DiaTool-DPO achieves highest Macro Avg. across all three model sizes
- DPO-only (without SFT) performs poorly — DPO complements, not replaces, SFT
- Slot accuracy consistently improves with DiaTool-DPO across all models
- LLaMA-3-8B gains are most dramatic: Slot +43.5%, Relevance +10.5%
- Near GPT-4o performance: 94.8% info-gathering, 91% tool call rejection

# LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



## Improvement Summary

### Slot Accuracy

0.639 → 0.917

+43.5%

### Relevance

0.826 → 0.913

+10.5%

Micro Avg: 0.844 → 0.905

Macro Avg: 0.816 → 0.904

# Key Takeaways

## RL-Based Dialogue Control for Production TA-LLMs

01

### MDP Formulation Enables Systematic Trajectory Pairing

Defining 5 internal dialogue states and 3 query types allows automatic generation of chosen/rejected trajectory pairs — eliminating the need for expensive human annotation.

02

### Automatic Dataset Construction is Language-Agnostic

The paired trajectory dataset was built in Korean but generalizes to English. This approach is applicable across languages for production TA-LLM deployments.

03

### DPO Complements — Not Replaces — SFT

DiaTool-DPO-only performs poorly. The best results come from SFT + DiaTool-DPO, confirming that preference optimization and supervised training are complementary.

04

### Slot-Filling and Tool Rejection See the Largest Gains

LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5% over SFT-only, reaching near GPT-4o conversational tool use on smaller open-source models.

05

### Opens Path to GPT-4o-Level Performance on Smaller Models

By combining SFT with DiaTool-DPO, open-source models achieve 94.8% information-gathering and 91% tool call rejection accuracy — comparable to GPT-4o.